

回归与相关分析:

直线回归与相关分析(基本统计方法)

多元线性回归分析

Logistic回归分析

多元线性回归分析:

Analyze → Regression

→ Linear regression

→ dependent: y

independent: x1-x4

→ ok

例15-1

未标准化的系数
和其标准误

标准化的系数

Coefficients^a

		Unstandardized Coefficients		Standardized Coefficients		
Model		B	Std. Error	Beta	t	Sig.
1	(Constant)	5.943	2.829		2.101	.047
	x1	.142	.366	.078	.390	.701
	x2	.351	.204	.309	1.721	.099
	x3	-.271	.121	-.339	-2.229	.036
	x4	.638	.243	.398	2.623	.016

a. Dependent Variable: y

整个方程的假设检验结果:

ANOVA^b

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	133.711	4	33.428	8.278	.000 ^a
	Residual	88.841	22	4.038		
	Total	222.552	26			

a. Predictors: (Constant), x4, x2, x3, x1

b. Dependent Variable: y

P<0.001

Model Summary

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.775 ^a	.601	.528	2.0095

a. Predictors: (Constant), x4, x2, x3, x1

复相关系数

决定系数

校正决定系数

剩余标准差

偏回归系数的假设检验结果:

Coefficients^a

Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.
		B	Std. Error	Beta		
1	(Constant)	5.943	2.829		2.101	.047
	x1	.142	.366	.078	.390	.701
	x2	.351	.204	.309	1.721	.099
	x3	-.271	.121	-.339	-2.229	.036
	x4	.638	.243	.398	2.623	.016

a. Dependent Variable: y

$P < 0.05$

逐步回归:

Variables Entered/Removed^a

Model	Variables Entered	Variables Removed	Method
1	x4	.	Stepwise (Criteria: Probability-of-F-to-enter $\leq$.100, Probability-of-F-to-remove $\geq$.150).
2	x1	.	Stepwise (Criteria: Probability-of-F-to-enter $\leq$.100, Probability-of-F-to-remove $\geq$.150).
3	x3	.	Stepwise (Criteria: Probability-of-F-to-enter $\leq$.100, Probability-of-F-to-remove $\geq$.150).
4	x2	.	Stepwise (Criteria: Probability-of-F-to-enter $\leq$.100, Probability-of-F-to-remove $\geq$.150).
5	.	x1	Stepwise (Criteria: Probability-of-F-to-enter $\leq$.100, Probability-of-F-to-remove $\geq$.150).

a. Dependent Variable: y

Coefficients^a

Model	Unstandardized Coefficients		Standardized Coefficients	t	Sig.
	B	Std. Error	Beta		
1 (Constant)	3.006	2.364		1.272	.215
x4	.978	.254	.610	3.845	.001
2 (Constant)	1.310	2.308		.568	.576
x4	.732	.259	.456	2.833	.009
x1	.678	.296	.369	2.290	.031
3 (Constant)	4.309	2.776		1.552	.134
x4	.635	.253	.396	2.507	.020
x1	.545	.293	.297	1.861	.076
x3	-.219	.122	-.274	-1.785	.088
4 (Constant)	5.943	2.829		2.101	.047
x4	.638	.243	.398	2.623	.016
x1	.142	.366	.078	.390	.701
x3	-.271	.121	-.339	-2.229	.036
x2	.351	.204	.309	1.721	.099
5 (Constant)	6.500	2.396		2.713	.012
x4	.663	.230	.413	2.880	.008
x3	-.287	.112	-.360	-2.570	.017
x2	.402	.154	.354	2.612	.016

a. Dependent Variable: y

Model Summary

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.610 ^a	.372	.347	2.3651
2	.696 ^b	.484	.441	2.1867
3	.740 ^c	.547	.488	2.0935
4	.775 ^d	.601	.528	2.0095
5	.773 ^e	.598	.546	1.9721

a. Predictors: (Constant), x4

b. Predictors: (Constant), x4, x1

c. Predictors: (Constant), x4, x1, x3

d. Predictors: (Constant), x4, x1, x3, x2

e. Predictors: (Constant), x4, x3, x2

Logistic回归分析:

Analyze → Regression → binary logistic

→ dependent: y

covariates: x1-x8

method: forward: LR

options: enter: 0.10 remove: 0.15

→ OK

SPSS中筛选自变量的方法:

Forward: 即表示逐步回归

三种方法: 选入时用记分检验, 剔除时
标准不同

conditional

LR 基于最大似然估计(最可靠)

Wald



1 : 序号 1

	序号	x1
1	1	
2	2	
3	3	
4	4	
5	5	
6	6	
7	7	
8	8	
9	9	
10	10	
11	11	
12	12	
13	13	
14	14	
15	15	
16	16	
17	17	
18	18	
19	19	

- Reports
- Descriptive Statistics
- Tables
- Compare Means
- General Linear Model
- Generalized Linear Models
- Mixed Models
- Correlate
- Regression
- Loglinear
- Classify
- Data Reduction
- Scale
- Nonparametric Tests
- Time Series
- Survival
- Multiple Response
- Missing Value Analysis...
- Complex Samples
- Quality Control
- ROC Curve...

- Linear...
- Curve Estimation...
- Binary Logistic...
- Multinomial Logistic...
- Ordinal...
- Probit...
- Nonlinear...
- Weight Estimation...
- 2-Stage Least Squares...
- Optimal Scaling...

x4	x5	x6	x7
1	0	0	
1	0	0	
1	0	0	
1	0	0	
			1
			0
			0
			0
			0
			0
			0
			0
			0
			0
			0
			0
			0
			0
			0
1	0	0	
1	0	0	
1	1	0	

序号

x1

x2

x3

x4

x5

x6

x7

x8

Dependent:

y

Block 1 of 1

Previous

Next

Covariates:

x4

x5

x6

x7

x8

Method:

Forward:LR

Selection Variable:

Rule...

ategorical...

Save...

Options...

OK

Paste

Reset

Cancel

Help

Variables in the Equation

		B	S.E.	Wald	df	Sig.	Exp(B)
Step 1 ^a	x6	2.826	1.095	6.657	1	.010	16.875
	Constant	-.523	.315	2.751	1	.097	.593
Step 2 ^b	x5	1.828	.680	7.227	1	.007	6.219
	x6	3.059	1.144	7.143	1	.008	21.303
	Constant	-1.281	.461	7.715	1	.005	.278
Step 3 ^c	x5	1.722	.714	5.814	1	.016	5.597
	x6	3.028	1.176	6.627	1	.010	20.656
	x8	1.663	.785	4.493	1	.034	5.277
	Constant	-2.359	.770	9.378	1	.002	.095
Step 4 ^d	x1	.924	.477	3.758	1	.053	2.519
	x5	1.496	.744	4.044	1	.044	4.464
	x6	3.135	1.249	6.303	1	.012	23.000
	x8	1.947	.847	5.289	1	.021	7.008
	Constant	-4.705	1.543	9.295	1	.002	.009

a. Variable(s) entered on step 1: x6.

b. Variable(s) entered on step 2: x5.

c. Variable(s) entered on step 3: x8.

d. Variable(s) entered on step 4: x1.

各自变量的
回归系数

如何解释? (X_6 :
动物脂肪摄入)

三、生存分析: survival

小样本:

Analyze → survival → kaplan meier

Time: time

Status: status(定义终点事件)

Factor: group

→ Compare factor levels: Test Statistics:

Log rank

→ Options: Plots: Survival

→ OK

例17-1

File	Edit	View	Data	Transform
				
14 :				
	time	stat		
1	1			
2	3			
3	5			
4	6			
5	7			
6	8			
7	10			
8	14			
9	17			
10	19			
11	20			
12	22			
13	26			
14	31			
15	34			
16	34			
17	44			
18	59			
19	1			

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- ROC Curve...

- Life Tables...
- Kaplan-Meier...
- Cox Regression...
- Cox w/ Time-Dep Cov...

用K-M法估计生存率

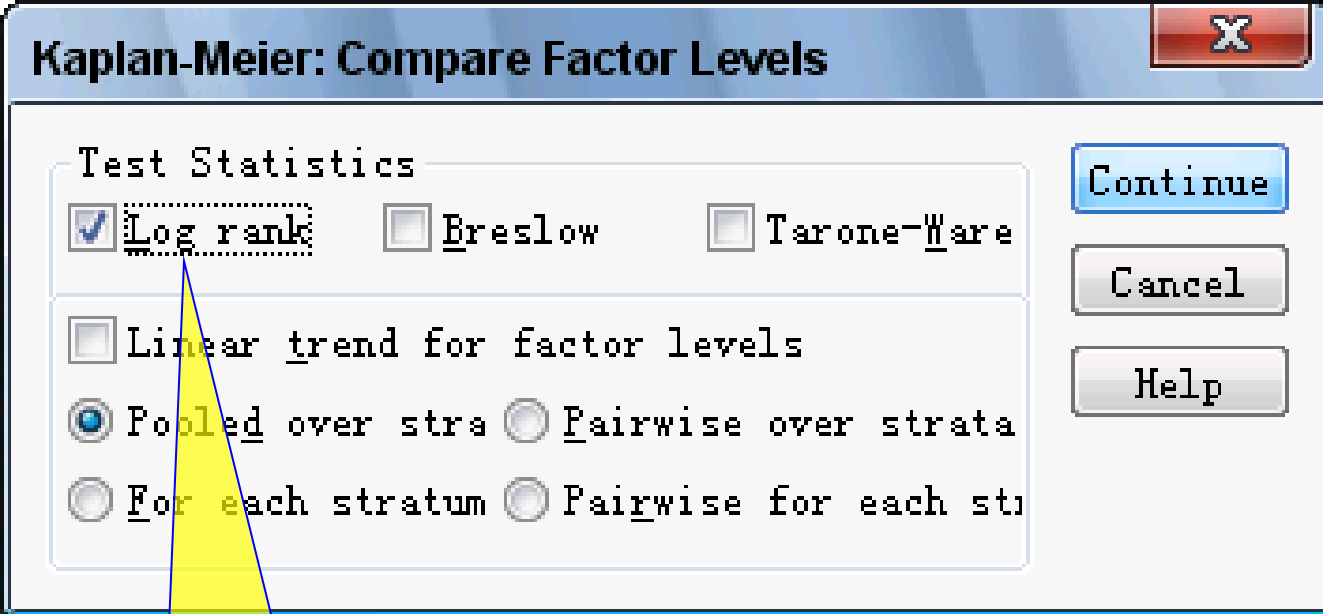


生存时间

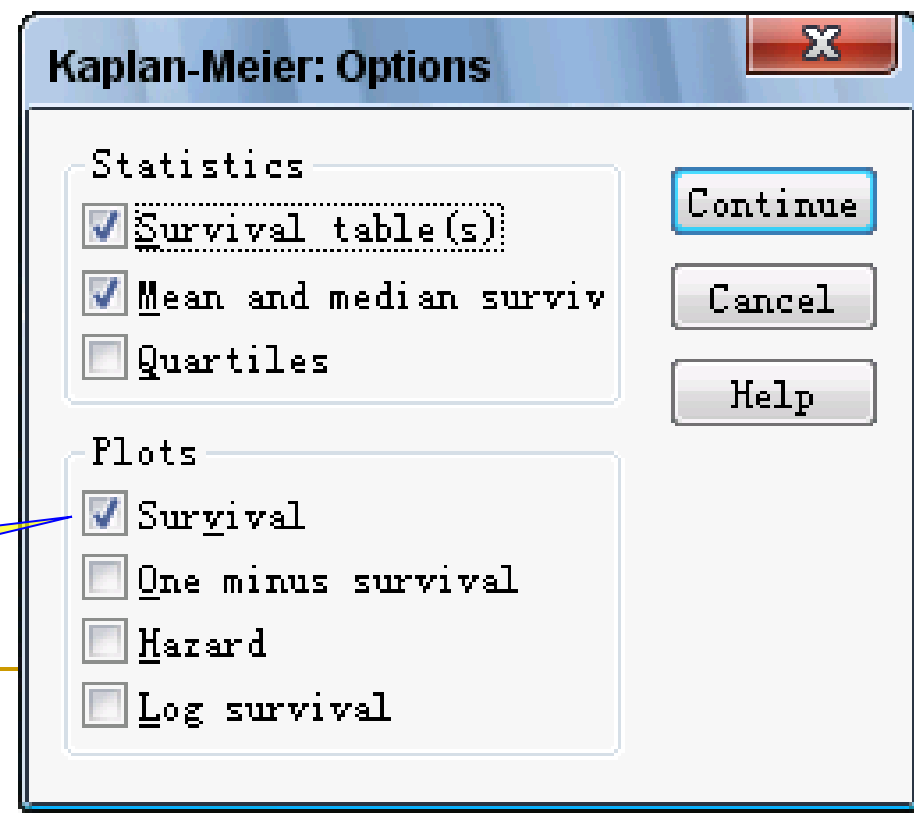
定义终点事件

选择Log-rank检验

欲比较的组别



选择Log-rank检验



绘制生存曲线

Survival Table

group		Time	Status	Cumulative Proportion Surviving at the Time		N of Cumulative Events	N of Remaining Cases
				Estimate	Std. Error		
1	1	1.000	1	.957	.043	1	22
	2	3.000	1	.913	.059	2	21
	3	5.000	1	.783	.086	5	18
	4	6.000	1	.652	.099	8	15
	5	7.000	1	.609	.102	9	14
	6	8.000	1	.565	.103	10	13
	7	10.000	1	.478	.104	12	11
	8	14.000	0	.	.	12	10
	9	17.000	1	.430	.104	13	9
	10	19.000	0	.	.	13	8
	11	20.000	0	.	.	13	7
	12	22.000	0	.	.	13	6
	13	26.000	0	.	.	13	5
	14	31.000	0	.	.	13	4
	15	34.000	1	.323	.122	14	3
	16	34.000	0	.	.	14	2
	17	44.000	1	.161	.129	15	1
	18	59.000	1	.000	.000	16	0
2	1	1.000	1	.900	.067	2	18
	2	2.000	1	.850	.080	3	17
	3	3.000	1	.750	.097	5	15

生存率及其标准误

Means and Medians for Survival Time

group	Mean ^a				Median			
	Estimate	Std. Error	95% Confidence Interval		Estimate	Std. Error	95% Confidence Interval	
			Lower Bound	Upper Bound			Lower Bound	Upper Bound
1	24.228	4.991	14.444	34.011	10.000	6.955	.000	23.632
2	7.800	1.176	5.496	10.104	6.000	2.981	.156	11.844
Overall	16.440	2.943	10.671	22.209	9.000	1.402	6.253	11.747

a. Estimation is limited to the largest survival time if it is censored.

平均及中位生存时间

生存率比较的
Log-rank检验

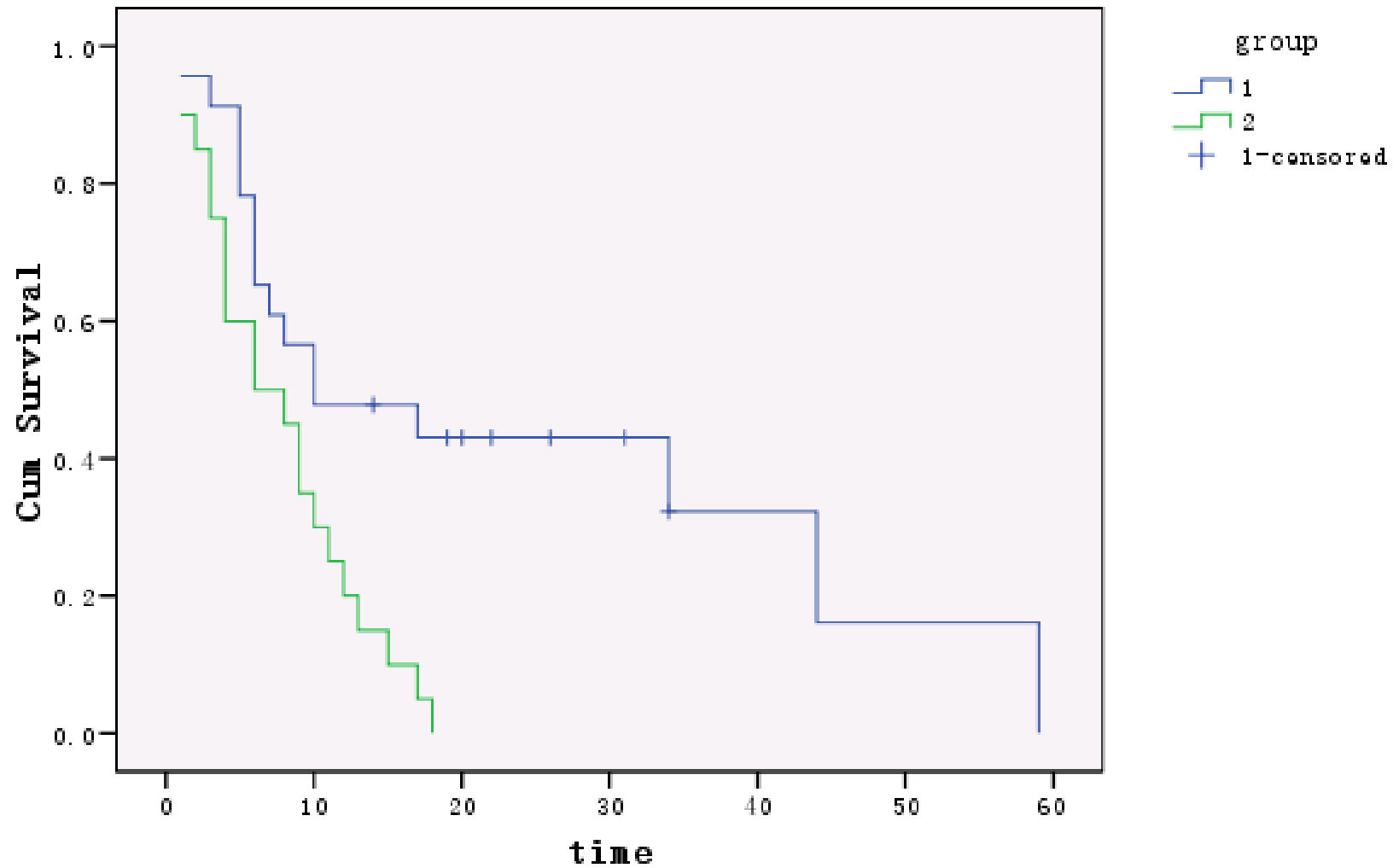
若假设检验有统计意义，可从以下几方面来评价各组效应大小：生存曲线图目测判断、中位生存时间比较等。

Overall Comparisons

	Chi-Square	df	Sig.
Log Rank (Mantel-Cox)	8.754	1	.003

Test of equality of survival distributions for the different levels of group.

Survival Functions



Cases weighted by f

大样本:

Data → weight cases:f

Analyze → survival→life tables

...

COX回归:

Analyze → Survival → Cox Regression:

Time: t (生存时间)

Status: $y(0)$ (终点事件)

Covariate: $x_1 x_2 x_3 x_4 x_5 x_6$ (影响因素)

→ OK

例17-5



	Name	Type	Width	Decimals	Label	Values	Mis
1	no	Numeric	8	0	编号	None	None
2	x1	Numeric	8	0	age	None	None
3	x2	Numeric	8	0	sex	{0, F}...	None
4	x3	Numeric	8	0	type	{0, 低分化}...	None
5	x4	Numeric	8	0	治疗方式	{0, old}...	None
6	x5	Numeric	8	0	淋巴转移	{0, N}...	None
7	x6	Numeric	8	0	浸润程度	{0, 未突破浆	None
8	t	Numeric	8	0	生存时间(m)	None	None
9	y	Numeric	8	0	结局	{0, 死亡}...	None
10							
11							
12							
13							
14							
15							
16							
17							

变量定义



1 : no	1
--------	---

	no	x1
1	1	5
2	2	5
3	3	5
4	4	4
5	5	4
6	6	4
7	7	4
8	8	3
9	9	3
10	10	4
11	11	4
12	12	4
13	13	5
14	14	5
15	15	4
16	16	5
17	17	4
18	18	4

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- Survival ▶
- Multiple Response ▶
- Missing Value Analysis... ▶
- Complex Samples ▶

Cox回归分析

- Life Tables...
- Kaplan-Meier...
- Cox Regression...
- Cox w/ Time-Dep Cov...

x4	x5	x6	t
0	1	0	
1	0	0	
1	1	1	
0	1	0	
1	0	2	
1	0	2	
1	0	2	
1	1	1	
0	0	1	
		1	
		0	
		2	
		0	
1	0	1	
0	0	1	
0	0	1	
0	0	2	
0	1	1	

Cox回归分析

Life Tables...

Kaplan-Meier...

Cox Regression...

Cox w/ Time-Dep Cov...

1

Cox Regression

Time: # 生存时间(m) [t]

Status: y(0)

Define Event...

Block 1 of 1

Previous Next

Covariates: x1 x2 x3 x4 x5

Method: Forward:LR

Strata:

OK Paste Reset Cancel Help

生存时间

定义终点事件

协变量（影响因素）

逐步回归法

6	
0	
0	
1	
0	
2	
2	
2	
1	
1	
1	
0	
2	
0	
1	
1	
1	
2	

17 17 48 1 1 0 0

进入方程中的自变量。
如何评价 x_4 、 x_5 对肿瘤
预后的影响？

Variables in the Equation

		B	SE	Wald	df	Sig.	Exp(B)
Step 1	x4	1.751	.546	10.265	1	.001	5.758
Step 2	x4	1.762	.548	10.337	1	.001	5.822
	x5	.931	.445	4.389	1	.036	2.538

Thank You